Complex Analysis Master Course

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Chapter 1

First Week

1.1 Lecture 1 12/10/2026

Definition 1. *the rectangular form of* $f : D \subseteq \mathbb{C} \to \mathbb{C}$ *is given by* $f(z) = u(x,y) + iv(x,y)$*, where* $z = x + iy$ *and* $u, v : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ *are real-valued functions.*

Example 1. *Consider the function* $f(z) = z^2$*. Then* $f(z) = (x+iy)^2 = x^2 - y^2 + 2ixy$*. Thus,* $u(x,y) = x^2 - y^2$ *and* $v(x,y) = 2xy$*.*

Definition 2. *the polar form of* $f : D \subseteq \mathbb{C} \to \mathbb{C}$ *is given by* $f(z) = re^{i\theta}$*, where* $z = re^{i\theta}$ *and* $r \geq 0$*,* $\theta \in [0, 2\pi)$*.*

Example 2. *Consider the function* $f(z) = z^2$*. Then* $f(z) = (re^{i\theta})^2 = r^2 e^{i2\theta}$*. Thus, the polar form of* $f(z)$ *is given by* $r^2 e^{i2\theta}$*.*

Definition 3. $\lim_{z \to z_0} f(z) = w_0$ *if for every* $\epsilon > 0$*, there exists a* $\delta > 0$ *such that* $|z - z_0| < \delta$ *implies* $|f(z) - w_0| < \epsilon$*.*

Remark 1. $\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$ *and* $\lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$ *if and only if* $\lim_{z \to z_0} f(z) = w_0$*.*

Definition 4. f *is continuous at* z_0 *if* $\lim_{z \to z_0} f(z) = f(z_0)$*.*

Definition 5. f *is differentiable at* z_0 *if the limit*

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists.

Example 3. *consder the function* $f(z) = |z|^2$ *use the definition to find the* $f'(0)$*.*

Solution 1. *We have* $f(z) = |z|^2 = z\bar{z}$. *Then*

$$f'(0) = \lim_{z\to 0} \frac{f(z) - f(0)}{z - 0} = \lim_{z\to 0} \frac{|z|^2 - 0}{z} = \lim_{z\to 0} \frac{z\bar{z}}{z} = \lim_{z\to 0} \bar{z} = \lim_{(x,y)\to(x_0,y_0)} u(x,y) - i \lim_{(x,y)\to(x_0,y_0)} v(x,y) = 0.$$

Thus, $f'(0) = 0$.

Theorem 1. *Let* $f(z) = u(x,y) + iv(x,y)$ *be a complex function. Then* f *is differentiable at* $z_0 = x_0 + iy_0$ *if and only if the partial derivatives of* u *and* v *exist and are continuous in a neighborhood of* (x_0, y_0)*, and satisfy the Cauchy-Riemann equations:*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Example 4. *Consider the function* $f(z) = z^2$. *Then* $f(z) = (x + iy)^2 = x^2 - y^2 + 2ixy$. *Thus,* $u(x,y) = x^2 - y^2$ *and* $v(x,y) = 2xy$. *We have*

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 2x, \quad \frac{\partial u}{\partial y} = -2y, \quad \frac{\partial v}{\partial x} = 2y.$$

Thus, the Cauchy-Riemann equations are satisfied and f *is differentiable at every point in* $\mathbb{C}$.

Example 5. *Consider the function* $f(z) = \bar{z}$. *Then* $f(z) = x - iy$. *Thus,* $u(x,y) = x$ *and* $v(x,y) = -y$. *We have*

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial y} = -1, \quad \frac{\partial u}{\partial y} = 0, \quad \frac{\partial v}{\partial x} = 0.$$

Thus, the Cauchy-Riemann equations are not satisfied and f *is not differentiable anywhere.*

Definition 6. *A function* f *is said to be an analytic at a point* z_0 *if it is differentiable in a neighborhood of* z_0. *If* f *is analytic at every point in a domain* D*, then* f *is said to be analytic in* D.

Example 6. *Consider the function* $f(z) = |z|^2$ *differentiable at* $z_0 = 0$ *but not analytic at* $z_0 = 0$ *since it is not differentiable in any neighborhood of* $z_0 = 0$.

Definition 7. *Consider The point* z_0 *is said to be a singular point of* f *if* f *is not analytic at* z_0*.and there exists a sequence of points* $z_n \to z_0$ *such that* f *is analytic at each* z_n.

Definition 8. z_0 *is said to be an isolated singular point of* f *if* f *is not analytic at* z_0 *and there exists a neighborhood of* z_0 *in which* f *is analytic at every point except* z_0.

Remark 2. *the neighborhood of* z_0 *in which* f *is analytic at every point except* z_0 *is called a punctured neighborhood of* z_0.

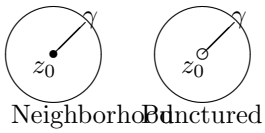


Figure 1.1: A neighborhood and a punctured neighborhood of z_0

Exercise 1. *1. Consider the function $f(z) = \frac{1}{z}$. Then f is not analytic at $z_0 = 0$ and there exists a punctured neighborhood of $z_0 = 0$ in which f is analytic at every point except $z_0 = 0$. Thus, $z_0 = 0$ is an isolated singular point of f.*

2. *Consider the function $f(z) = \frac{1}{z^2+1}$. Then f is not analytic at $z_0 = i$ and $z_0 = -i$ and there exists a punctured neighborhood of $z_0 = i$ and $z_0 = -i$ in which f is analytic at every point except $z_0 = i$ and $z_0 = -i$.*

3. *Consider the function $f(z) = \frac{1}{z^2}$. Then f is not analytic at $z_0 = 0$ and there exists a punctured neighborhood of $z_0 = 0$ in which f is analytic at every point except $z_0 = 0$. Thus, $z_0 = 0$ is an isolated singular point of f.*

4. *Consider the function $f(z) = \frac{1}{z^2+1}$. Then f is not analytic at $z_0 = i$ and $z_0 = -i$ and there exists a punctured neighborhood of $z_0 = i$ and $z_0 = -i$ in which f is analytic at every point except $z_0 = i$ and $z_0 = -i$. Thus, $z_0 = i$ and $z_0 = -i$ are isolated singular points of f.*

5. *Consider the function $f(z) = \frac{1}{\sin(\pi/z)}$. Its singularities at $z = 1/n$, $n \in \mathbb{Z} \setminus \{0\}$, accumulate at 0. Thus, $z = 0$ is a non-isolated singular point.*

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Example 7. *$f(z) = \frac{1}{\sin(\pi/z)}$. Determine whether $z = 0$ is a singular point (isolated or not).*

Solution 2. *The function is not defined at $z = 0$. To find its other singularities, solve*

$$\sin\left(\frac{\pi}{z}\right) = 0 \quad \Longleftrightarrow \quad \frac{\pi}{z} = n\pi, \qquad n \in \mathbb{Z}.$$

Since $z \neq 0$, the solutions are $z = 1/n$ for $n \in \mathbb{Z} \setminus \{0\}$. These are poles of f, and they accumulate at 0 because $1/n \to 0$ as $|n| \to \infty$. Consequently, every punctured neighborhood of 0 contains singularities of f. Therefore, $z = 0$ is a non-isolated singular point, not an isolated singular point.

Example 8. *$f(z) = \log z$. Determine whether $z = 0$ is an isolated singularity.*

Solution 3. *The complex logarithm is multivalued: if $z = re^{i\theta}$, then*

$$\log z = \ln r + i(\theta + 2\pi k), \qquad k \in \mathbb{Z}.$$

When z travels once around 0, its argument increases by 2π, so the value of the logarithm changes by $2\pi i$. Thus, no single-valued analytic branch of $\log z$ exists on a full punctured neighborhood of 0. The point $z = 0$ is a branch point (often called a logarithmic branch point), not an isolated singularity in the sense requiring analyticity throughout a punctured neighborhood. A branch such as the principal logarithm is analytic only after a branch cut is removed.

Example 9. *Let $f(z) = \operatorname{Log} z$ be the principal logarithm. Determine whether $z = 0$ is an isolated singularity.*

Solution 4. *The principal logarithm is defined by*

$$\operatorname{Log} z = \ln|z| + i \operatorname{Arg} z, \qquad -\pi < \operatorname{Arg} z < \pi,$$

on the slit plane $\mathbb{C} \setminus (-\infty, 0]$. It is analytic throughout this domain. However, every disk centered at 0 meets the branch cut, so there is no full punctured neighborhood of 0 on which this branch is defined and analytic. Therefore, $z = 0$ is not an isolated singularity of the principal logarithm in the standard sense; it is a branch point. As $z \to 0$ within the slit plane, $\operatorname{Re}(\operatorname{Log} z) = \ln|z| \to -\infty$.

1.3 Summary of Examples: Singularities

Remark 3. *An isolated singularity requires the function to be analytic on a full punctured neighborhood of the point. A branch point such as 0 for $\log z$ is therefore not an isolated singularity in this sense.*

1.4 Integration

Definition 9. *Let $z : [a, b] \to \mathbb{C}$ where $z(t) = x(t) + iy(t)$ Then $\int_a^b z(t)dt$*

Table 1.1: Summary of the functions discussed and their singularity behavior.

Function	Point(s)	Classification	Explanation
z^2	None	No singularities; entire	It is analytic at every point of $\mathbb{C}$.
$\bar{z}$	Every point	Not analytic anywhere	It fails the Cauchy–Riemann equations everywhere, so there is no punctured neighborhood on which it is analytic.
$\lvert z\rvert^2$	0	Differentiable at 0, but not an isolated singularity	It is differentiable at 0 but not analytic in any neighborhood of 0.
$\frac{1}{z}$	0	Isolated pole (order 1)	It is analytic in a punctured neighborhood of 0.
$\frac{1}{z^2}$	0	Isolated pole (order 2)	It is analytic in a punctured neighborhood of 0.
$\frac{1}{z^2+1}$	$i, -i$	Isolated simple poles	The denominator has simple zeros at i and $-i$; each point has a punctured neighborhood containing no other singularities.
$\frac{1}{\sin(\pi/z)}$	$z = 1/n$, $n \neq 0$; accumulation at 0	Each $1/n$ is an isolated simple pole; 0 is a non-isolated singular point	The poles $1/n$ accumulate at 0, so no punctured neighborhood of 0 is free of other singularities.
$\log z$	0	Logarithmic branch point; not an isolated singularity	Analytic continuation once around 0 changes the logarithm by $2\pi i$, so no single-valued analytic branch exists on a full punctured neighborhood.
$\operatorname{Log} z$ (principal branch)	0	Branch point; not an isolated singularity	The principal branch is analytic on the slit plane $\mathbb{C} \setminus (-\infty, 0]$, not on a full punctured neighborhood of 0.